

Indian School Performance Analysis

A Project Report

Submitted by

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Declaration

I, **Jasmeen** , hereby declare that this project report titled "**Indian School Performance Analysis**" is my original work carried out as a part of my B.Tech in Artificial Intelligence and Machine Learning. This project has been completed under the guidance of **Mr. Harkirat Singh**.

I confirm that this project has not been submitted to any other university or institution for the award of any degree or certificate. The data used in this project have been collected from reliable sources.

Place :

signature of student

Date :

Acknowledgment

I would like to express my sincere gratitude to everyone who supported and guided me throughout the completion of this project. This project has been a valuable learning experience, and I am thankful to all those who helped me during this journey. First of all, I would like to thank my project guide and all the faculty members of the Artificial Intelligence and Machine Learning Department for their valuable guidance, encouragement, and continuous support. Their suggestions and feedback helped me improve my work and complete this project successfully. I am also thankful to my college for providing the resources and learning environment that helped me gain knowledge and develop the skills required for this project. The concepts I learned during my course played an important role in completing this work. I would also like to thank my family for their constant support, motivation, and encouragement. Their patience and belief in me gave me the confidence to complete this project with dedication.

Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of this project. This project has helped me improve my technical knowledge, problem-solving skills, and understanding of data analysis and visualization. I am grateful for the opportunity to work on this project and learn from this experience.

Bonafide Certificate

Abstract

This project, **Indian School Performance Analysis**, aims to analyze different aspects of school education in India using data related to dropout rates, gross enrollment ratio (GER), and school infrastructure. The main objective is to understand how educational facilities and enrollment are related to student dropout rates across different states. The data was cleaned, processed, and analyzed using Python libraries, and an interactive dashboard was developed using Streamlit to present the results through various charts and graphs. The project helps identify trends, compare the performance of different states, and provide useful insights that can support better educational planning and decision-making. Overall, this project demonstrates how data analysis and visualization can be used to better understand the current state of school education in India.

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CHAPTER-1

INTRODUCTION

1.1 INTRODUCTION

Education plays an important role in the growth of individuals and the development of a nation. A strong education system helps students gain knowledge, improve their skills, and build a better future. In India, school education is provided through government and private institutions across different states. Although many improvements have been made over the years, several challenges such as dropout rates, unequal educational facilities, and differences in enrollment still affect the quality of education.

Today, educational data is collected from various schools and states. This data can be analyzed to understand the current condition of the education system. By using data analysis and visualization techniques, it becomes easier to identify trends, compare the performance of different states, and support better decision-making.

The **Indian School Performance Analysis** project focuses on analyzing school education data related to dropout rates, Gross Enrollment Ratio (GER), and school infrastructure. The project uses Python and Streamlit to develop an interactive dashboard that presents the analysis through different charts and graphs, making the information easy to understand.

1.2 Background of study

India has one of the largest school education systems in the world. Every year, millions of students enroll in schools, but the quality of education varies across different states. Some schools have better infrastructure and learning facilities, while others still lack basic resources such as classrooms, electricity, drinking water, toilets, and computers.

Government initiatives have improved education in many areas, but dropout rates and unequal access to facilities remain major concerns. These challenges highlight the need to study educational data and understand how different factors influence school performance.

This project combines data related to infrastructure, Gross Enrollment Ratio (GER), and dropout rates to provide a better understanding of the current state of school education in India.

1.3 Problem statement

Educational information is available in different datasets, making it difficult to analyze all important factors together. As a result, understanding the relationship between school infrastructure, enrollment, and dropout rates becomes challenging. There is a need for a system that combines these datasets and presents the information in a simple and interactive way. This project addresses this need by developing a dashboard that allows users to explore educational data through charts, graphs, and state-wise comparisons.

1.4 Objectives

The main objectives of this project are:

- To analyze school education data from different states of India.
- To study dropout rates, Gross Enrollment Ratio (GER), and school infrastructure.
- To clean and preprocess educational datasets.
- To merge multiple datasets for analysis.
- To compare educational performance across different states.
- To identify trends and patterns in educational data.
- To create interactive charts and visualizations.
- To develop a user-friendly dashboard using Streamlit.
- To provide useful insights for better educational planning.

1.5 Scope of Project

The project focuses on analyzing school education data available in the selected datasets. It mainly studies three important factors: dropout rates, Gross Enrollment Ratio (GER), and school infrastructure. The analysis is performed at the state level to compare educational performance across India.

The project provides an interactive dashboard that allows users to explore the data using charts and filters. However, it does not predict future outcomes or use machine learning models. The analysis is limited to the available historical data.

1.6 Significance of the Project

This project helps users understand the current condition of school education in India through data analysis and visualization. It allows easy comparison between states and highlights areas that require improvement.

The dashboard can be useful for students, researchers, educators, and policymakers. It also demonstrates how data analytics can be applied to solve real-world problems in the education sector.

1.7 Project overview

The Indian School Performance Analysis project is developed using Python and Streamlit. It combines three educational datasets related to dropout rates, Gross Enrollment Ratio (GER), and school infrastructure. After preprocessing the data, different visualizations are created to present meaningful information.

The dashboard includes multiple analysis sections, interactive filters, and charts that help users explore educational trends. It provides a simple and effective way to understand school performance across different states of India.

Chapter 2

Literature Review

2.1 Introduction

A literature review is an important part of any research project because it helps understand the work that has already been done in a particular field. It provides information about existing studies, methods, and findings related to the project topic. It also helps identify research gaps and explains how the current project is different from previous work. For this project, the literature review focuses on educational data analysis, school infrastructure, Gross Enrollment Ratio (GER), dropout rates, and the use of data visualization in education. These topics provide the foundation for understanding the importance of analyzing school performance in India.

2.2 Educational Data Analysis

Educational institutions generate a large amount of data every year, including information about student enrollment, attendance, school facilities, examination results, and dropout rates. Analyzing this data helps in understanding educational trends and identifying areas that require improvement.

Data analysis makes it easier to compare educational performance across different states and districts. It also helps decision-makers identify schools that need additional resources or support. With the help of modern programming tools and visualization techniques, educational data can be converted into meaningful information that is easy to understand.

Educational data analysis is becoming increasingly important because it supports evidence-based planning and allows governments and educational institutions to make better decisions.

2.3 Indian School Education system

India has one of the largest education systems in the world, with millions of students enrolled in government and private schools. The school education system includes primary, upper primary, secondary, and higher secondary levels. Over the years, several government initiatives have been introduced to improve enrollment and reduce dropout rates.

Although significant progress has been made, educational performance still differs among states. Some states have better educational facilities and higher enrollment, while others continue to face challenges related to infrastructure and student retention. These differences make educational data analysis important for understanding the current status of school education across the country.

2.4 School Infrastructure

School infrastructure plays an important role in providing a positive learning environment for students. Basic facilities such as classrooms, electricity, drinking water, toilets, libraries, playgrounds, and computers contribute to better educational experiences.

Schools with good infrastructure generally provide a safer and more comfortable environment for students. These facilities encourage regular attendance and support effective teaching and learning. On the other hand, schools with poor infrastructure may face difficulties in maintaining student interest and participation.

In this project, different infrastructure facilities are combined to calculate an Infrastructure Score, which helps compare the overall quality of school facilities across different states.

2.5 Gross Enrollment Ratio

Gross Enrollment Ratio (GER) is an important indicator used to measure participation in education. It represents the total number of students enrolled at a particular level of education as a percentage of the population in the corresponding age group.

A higher GER generally indicates better access to education and greater participation of students in schools. However, GER alone does not measure the quality of education. Therefore, it should be analyzed together with other indicators such as dropout rates and school infrastructure.

By studying GER, policymakers can identify regions where educational participation needs improvement and develop suitable educational programs.

2.6 Dropout Rate

The dropout rate refers to the percentage of students who leave school before completing a particular level of education. It is considered one of the major challenges faced by the education system.

Students may leave school for various reasons, including financial difficulties, family responsibilities, lack of educational facilities, long travel distances, health issues, or limited access to quality teaching. High dropout rates can negatively affect literacy, employment opportunities, and overall social development.

Analyzing dropout data helps identify states where students are more likely to discontinue their education and supports the development of strategies to improve student retention.

2.7 Importance of Data Visualization

Data visualization is the process of presenting information using charts, graphs, maps, and dashboards. It makes large datasets easier to understand by showing trends, comparisons, and patterns in a visual form.

In educational analysis, visualizations help users quickly identify differences between states, compare educational indicators, and understand relationships among multiple variables. Interactive dashboards further improve the user experience by allowing users to explore data through filters and different chart types.

This project uses an interactive Streamlit dashboard to present educational information in a simple and user-friendly manner.

2.8 Reasearch Gap

Although several studies have examined different aspects of school education, many of them analyze infrastructure, enrollment, or dropout rates separately. This makes it difficult to understand the relationship between these important educational indicators.

Another limitation is that many studies present their findings using static tables and reports, which are less interactive and harder to explore. There is a need for a system that combines multiple educational datasets and presents the information through interactive visualizations.

This project addresses these gaps by integrating infrastructure, Gross Enrollment Ratio, and dropout rate datasets into a single dashboard. Users can compare states, explore trends, and gain meaningful insights through interactive charts and filters.

2.9 Summary

The literature review shows that educational data analysis plays an important role in understanding the performance of the school education system. School infrastructure, Gross Enrollment Ratio, and dropout rates are key indicators that help evaluate educational development. Previous studies have contributed valuable knowledge, but there is still a need for integrated and interactive analysis tools.

The Indian School Performance Analysis project attempts to fill this gap by combining multiple educational indicators into a single Streamlit dashboard. The project provides meaningful visualizations and insights that can support researchers, students, educators, and policymakers in understanding educational performance across different states of India.

Chapter 3

System Analysis and Methodology

3.1 Introduction

This chapter explains the complete methodology followed to develop the **Indian School Performance Analysis** project. It describes the system requirements, datasets used, data collection process, preprocessing techniques, and the workflow of the project. The main purpose of this chapter is to explain how raw educational data was converted into meaningful information through data analysis and visualization. The project analyzes educational data related to school infrastructure, Gross Enrollment Ratio (GER), and dropout rates across different states of India. The data is processed and presented through an interactive dashboard to help users understand educational trends and compare state-wise performance.

3.2 System requirements

The project was developed using a personal computer with standard hardware and software requirements. Python was used as the programming language because it provides powerful libraries for data analysis and visualization. Streamlit was used to create the interactive dashboard, while Pandas and NumPy were used for data manipulation. Plotly was used to generate interactive

Hardware Requirement

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB (8 GB Recommended)
- Storage: At least 20 GB Free Space
- Operating System: Windows 10 or Windows 11

Software Requirement

- Python 3.x
- Streamlit
- Pandas
- NumPy
- Plotly
- Visual Studio Code
- Web Browser (Google Chrome)

3.3 Dataset Requirement

The project uses educational data containing information about school infrastructure, Gross Enrollment Ratio (GER), and dropout rates. These data sources provide valuable information for analyzing the performance of school education across different states of India.

Infrastructure Dataset

The infrastructure dataset contains information about the availability of basic facilities in schools. It includes data such as electricity, drinking water, boys' toilets, girls' toilets, computers, libraries, and other educational resources. An infrastructure score was calculated using these facilities to compare different states.

Gross Enrollment Ratio (GER) Dataset

The GER dataset provides information about the percentage of students enrolled in schools compared to the eligible school-age population. It helps measure the level of participation in education across different states.

Dropout Rate Dataset

The dropout dataset contains information about students who leave school before completing their education. It helps identify states where dropout rates are higher and allows comparisons with infrastructure and enrollment levels.

3.4 Data Collection

The datasets used in this project were collected from publicly available educational data sources. After downloading the datasets, they were carefully examined to understand their structure, columns, and data types. Only the required features were selected for analysis to improve the efficiency of the project.

3.5 Data Preprocessing

Before performing analysis, the datasets were cleaned and preprocessed to improve data quality. Missing values and unnecessary columns were removed wherever required. Column names were renamed to maintain consistency across all datasets. Data types were converted into appropriate formats, and duplicate records were checked to avoid incorrect results.

The data was cleaned and prepared for analysis by removing unnecessary columns, handling missing values, renaming columns where required, checking for duplicate records, and converting data into appropriate formats. An infrastructure score was calculated using the available school facilities to support meaningful analysis.. An infrastructure score was calculated using available school facilities to represent the overall quality of school infrastructure in each state. After preprocessing, the final dataset became suitable for visualization and analysis.

The preprocessing stage ensured that the dashboard displayed reliable and meaningful information.

3.6 Methodology

The project follows a systematic approach to analyze educational data. First, the datasets were collected and imported into Python. Next, the data was cleaned and preprocessed by removing unnecessary

information, handling missing values, converting data types, and preparing it for analysis. Different visualizations were then created to study educational trends and compare the performance of different states.

After preparing the final dataset, different charts and graphs were created to study educational trends across Indian states. Interactive filters were added to the Streamlit dashboard so that users could explore the data according to their requirements. Finally, important insights were generated from the analysis, helping users understand the relationship between infrastructure, enrollment, and dropout rates.

3.7 Workflow of the System

The workflow of the project follows a sequence of steps from data collection to dashboard development. Each stage contributes to producing accurate and meaningful educational insights.

Workflow Steps

1. Data Collection
2. Data Cleaning
3. Data Preprocessing
4. Infrastructure Score Calculation
5. Data Analysis
6. Data Visualization
7. Interactive Dashboard Development
8. Key Insights

3.8 Dashboard Modules

The dashboard has been divided into different modules to make navigation simple and user-friendly.

Home Page

The home page provides an overview of the project and introduces users to the objectives and features of the dashboard.

Dataset Summary

This section presents a summary of the datasets used in the project, including the number of records, columns, and important educational indicators.

Dropout Rate Analysis

This module allows users to analyze dropout rates across different states using interactive charts. Users can compare state-wise performance and identify regions with higher dropout rates.

Gross Enrollment Ratio Analysis

This section displays the enrollment trends of different states through visualizations. It helps users understand the participation of students in school education.

Infrastructure Analysis

The infrastructure module evaluates school facilities available in different states. It compares infrastructure scores and highlights states with better educational resources.

Key Insights

This module summarizes the most important findings obtained from the analysis. It provides meaningful observations that help users understand the overall condition of school education in India.

Chapter- 4

System Design and Implementation

4.1 Introduction

This chapter explains the design and implementation of the **Indian School Performance Analysis** dashboard. The dashboard was developed using **Python** and **Streamlit** to provide an interactive platform for analyzing school education data in India. It presents information related to school infrastructure, Gross Enrollment Ratio (GER), and dropout rates through different graphs and visualizations. The main objective of the dashboard is to help users understand educational trends and compare the performance of different states in a simple and interactive way.

The dashboard is designed with a user-friendly interface that allows users to navigate through different sections using the sidebar menu. Each section focuses on a specific part of the project, making the analysis easy to understand even for users with little technical knowledge.

4.2 System Design

The dashboard follows a simple and organized design. The sidebar is used for navigation, while the main screen displays project information, statistics, graphs, and visualizations. The interface is divided into different modules so that users can access different analyses without confusion.

The overall design focuses on readability, simplicity, and easy navigation. Blue and dark color combinations are used to provide a professional appearance while maintaining good visibility of the dashboard components.

The dashboard contains four main modules:

- Overview
- Dataset Summary
- Data Cleaning
- Data Visualization

Each module performs a specific function and contributes to the complete analysis of Indian school education data.

4.3 Overview Module

The Overview page acts as the landing page of the dashboard. It introduces the project and provides a summary of the educational datasets used for analysis. The page displays the project title along with a

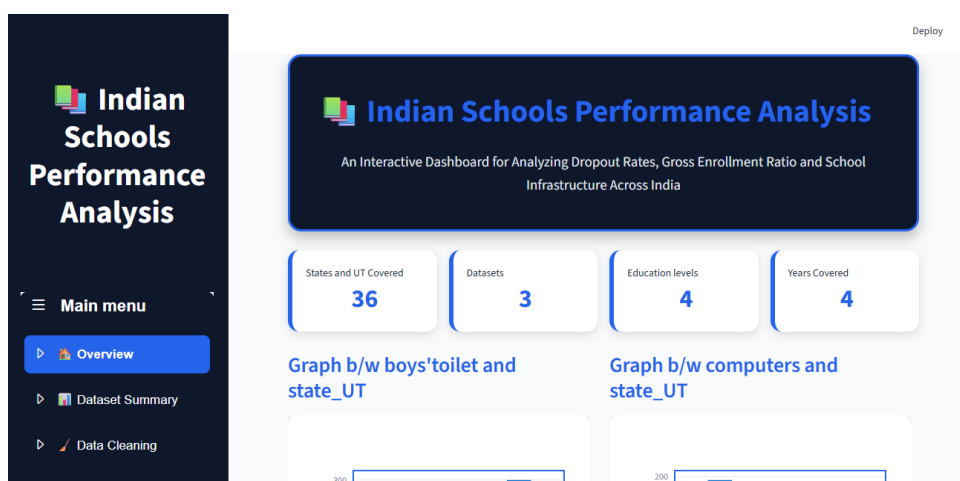
short description explaining that the dashboard analyzes dropout rates, Gross Enrollment Ratio, and school infrastructure across India.

To provide a quick understanding of the dataset, four information cards are displayed at the top of the page. These cards show important statistics such as:

- Number of States and Union Territories Covered
- Total Number of Datasets
- Number of Education Levels
- Number of Academic Years Covered

These summary cards help users understand the scope of the project before exploring the detailed analysis.

The lower section of the page displays sample graphs from different datasets. These graphs provide users with an initial understanding of the available educational information and encourage further exploration through the remaining dashboard modules.



4.4 Dataset Summary Module

The Dataset Summary module provides an overview of the educational data used in the project. It helps users understand the structure of the dataset by displaying information such as the number of rows, columns, important variables, and sample records.

This module allows users to become familiar with the available educational indicators before proceeding with further analysis. Displaying the dataset summary improves transparency and helps users understand the information used for generating the visualizations.

The summary also makes it easier to identify the major variables related to school infrastructure, enrollment, and dropout rates.

Dataset Preview

Rows to Preview 10

	State_UT	year	Primary_Boys	Primary_Girls	Primary_Total	Upper Primary_Boys	Upper Primary_Girls	Upper Pr
0	A & N Islands	2012-13	0.83	0.51	0.68	3.49	1.09	
1	A & N Islands	2013-14	1.35	1.06	1.21	3.49	1.54	
2	A & N Islands	2014-15	0.47	0.55	0.51	1.44	1.95	
3	Andhra Pradesh	2012-13	3.3	3.05	3.18	3.21	3.51	
4	Andhra Pradesh	2013-14	4.31	4.39	4.35	3.46	4.12	
5	Andhra Pradesh	2014-15	6.57	6.89	6.72	5.09	5.32	
6	Arunachal Pradesh	2013-14	11.54	10.22	10.89	4.44	6.74	
7	Arunachal Pradesh	2012-13	15.84	14.44	15.16	5.86	9.06	
8	Arunachal Pradesh	2014-15	11.51	10.09	10.82	5.31	8.08	
9	Assam	2012-13	7.02	5.46	6.24	7.89	6.55	

4.5 Data Cleaning Module

The Data Cleaning module explains the preprocessing steps performed before analysis. Since raw educational data may contain missing values, duplicate records, or unnecessary information, preprocessing is an important step in improving data quality.

The cleaning process involved:

- Removing unnecessary columns
- Checking for missing values
- Renaming columns where required
- Converting data into suitable formats
- Verifying the consistency of the dataset

These preprocessing steps ensure that the analysis is performed on accurate and well-organized data, resulting in reliable visualizations and insights.

After Cleaning ⇄

Original columns
14

Rows
110

Missing values
0

Duplicate
0

Rows 5

	State_UT	year	Primary_Boys	Primary_Girls	Primary_Total	Upper Primary_Boys	Upper Primary_Girls	Upper Pr
0	A & N Islands	2012-13	0.83	0.51	0.68	3.49	1.09	
1	A & N Islands	2013-14	1.35	1.06	1.21	3.49	1.54	
2	A & N Islands	2014-15	0.47	0.55	0.51	1.44	1.95	
3	Andhra Pradesh	2012-13	3.3	3.05	3.18	3.21	3.51	
4	Andhra Pradesh	2013-14	4.31	4.39	4.35	3.46	4.12	

4.6 Data Visualization Module

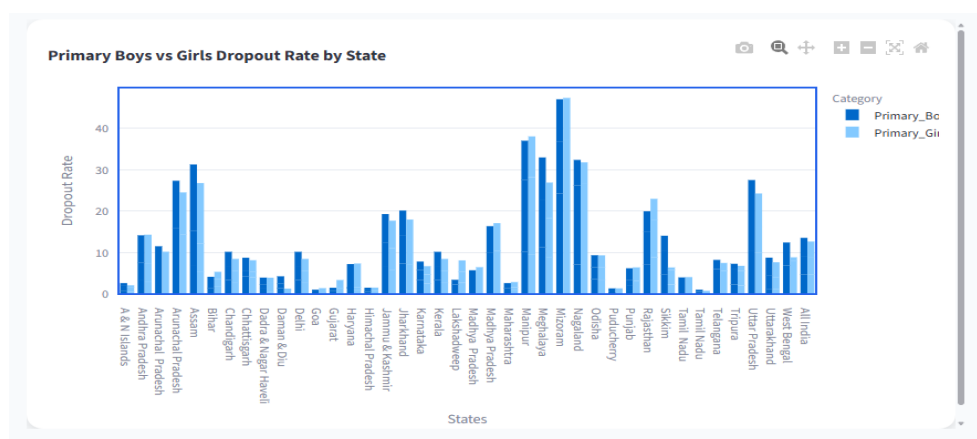
The Data Visualization module is the most important part of the dashboard. It presents educational information using interactive charts that help users compare different states and understand educational trends.

Visual representations make large amounts of educational data easier to interpret than numerical tables. Users can observe patterns, compare values, and identify differences among states through various graphical representations.

The dashboard contains visualizations related to:

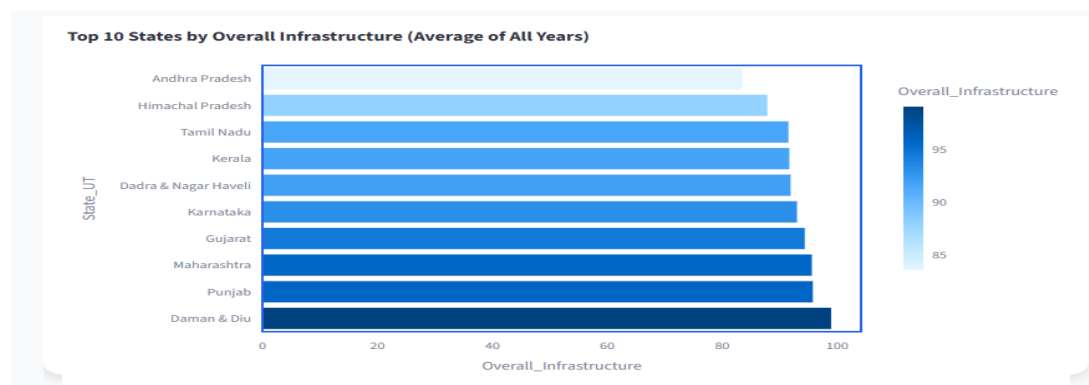
- School Infrastructure
- Gross Enrollment Ratio (GER)
- Dropout Rate

Each graph is designed to highlight a specific educational indicator and support meaningful analysis.

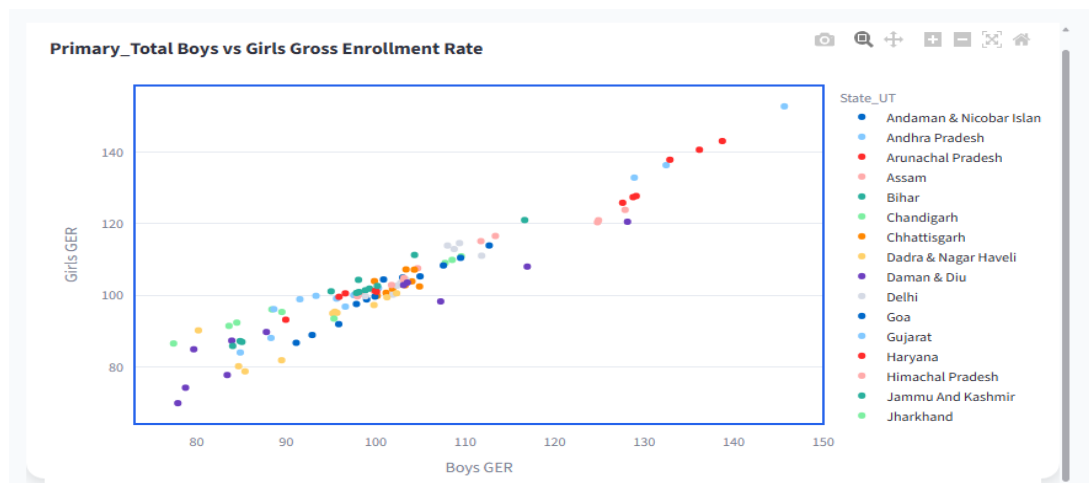


4.7 Graph Analysis

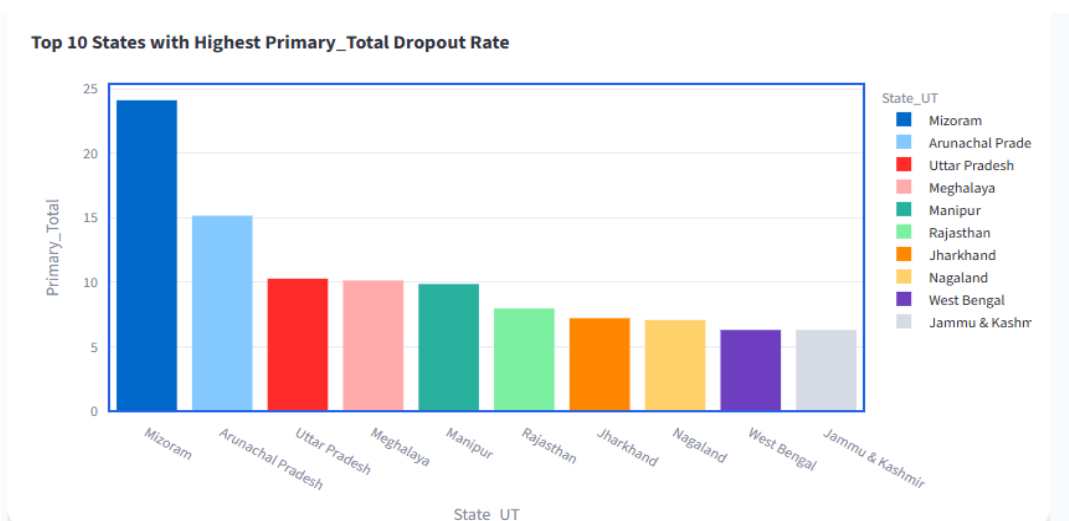
The first graph represents information from the school infrastructure dataset. It compares the availability of educational facilities among different states. This graph helps users understand how infrastructure varies across India and identifies states with better educational resources.



The second graph represents data related to Gross Enrollment Ratio (GER). It allows users to compare enrollment levels among different states and educational levels. The graph helps identify regions with higher or lower student enrollment and provides useful information about educational participation.



The third graph represents dropout rate data. It compares dropout rates across different states and helps identify regions where students are leaving school before completing their education. The graph makes it easier to recognize educational challenges that require further attention.

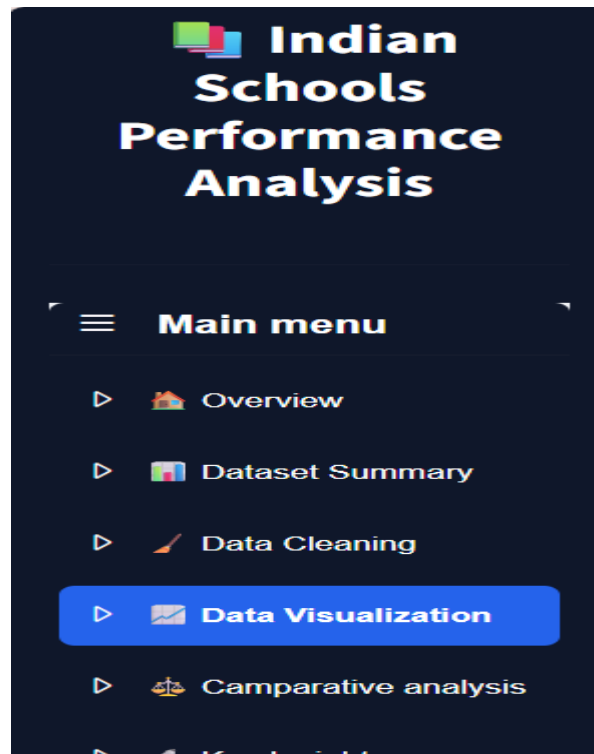


4.8 User Interface Design

The user interface was designed to provide a simple and interactive experience. A sidebar navigation menu is placed on the left side of the dashboard, allowing users to switch between different modules without difficulty.

The interface uses dark and blue color combinations to improve readability and maintain a consistent appearance throughout the dashboard. Information cards are displayed at the top of the Overview page to summarize important statistics, while graphs are arranged below them in a clear and organized manner.

The responsive layout ensures that users can access information without unnecessary scrolling or complex navigation.



4.9 Features of the Dashboard

The main features of the dashboard include:

- Interactive and user-friendly interface.
- Easy navigation through sidebar menu.
- Overview of educational statistics.
- Dataset summary for better understanding.
- Data cleaning information.
- Interactive graphs and visualizations.
- State-wise educational analysis.
- Simple presentation of educational indicators.
- Professional dashboard design.
- Easy interpretation of educational trends.

Chapter 5

Analysis and Visualization

5.1 Introduction

This chapter presents the results obtained from the analysis of the educational data through the **Indian School Performance Analysis** dashboard. The results are displayed using different graphs and visualizations, making it easier to understand educational trends across different states of India. The dashboard helps users compare school infrastructure, Gross Enrollment Ratio (GER), and dropout rates using interactive visualizations. The observations presented in this chapter are based on the analysis performed through the developed dashboard

5.2 Overview of Results

The dashboard provides a clear and organized view of educational data collected from different states of India. It allows users to compare educational indicators and identify variations in school performance. Through visual representations, users can quickly understand differences in infrastructure, enrollment, and dropout rates among different regions.

The dashboard also simplifies the interpretation of educational data by presenting important statistics and graphical comparisons on a single platform. This makes the analysis more effective than viewing large numerical tables.

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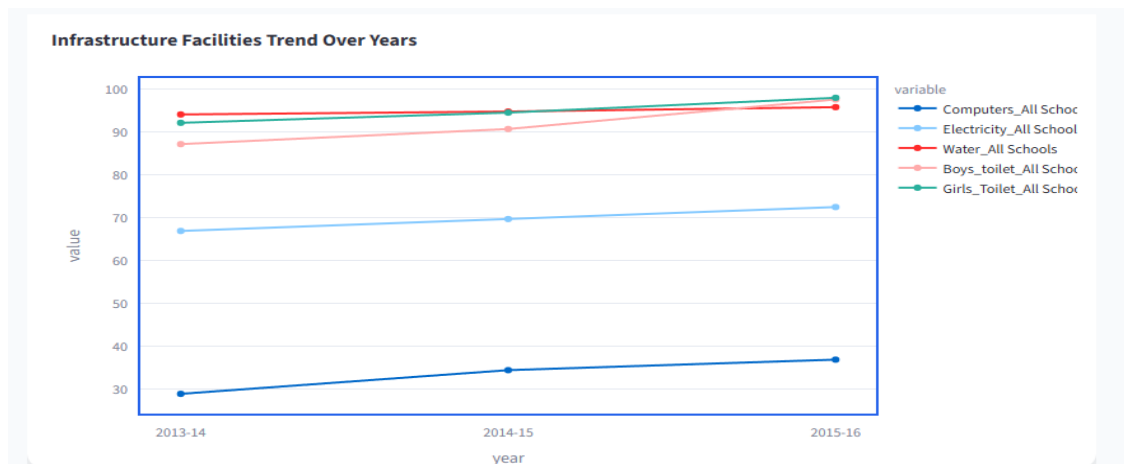
5.3 Results of School Infrastructure Analysis

The infrastructure analysis provides information about the availability of educational facilities in different states. The graphs show variations in facilities such as toilets, computers, electricity, drinking water, and other school resources.

From the analysis, it can be observed that some states have better educational infrastructure than others. Better infrastructure generally provides a more suitable learning environment for students and improves the quality of education. The visualizations also help identify states that may require additional improvements in school facilities.

Observation:

- Infrastructure differs from one state to another.
- Some states have better educational facilities than others.
- Infrastructure analysis helps identify areas where improvements are needed.



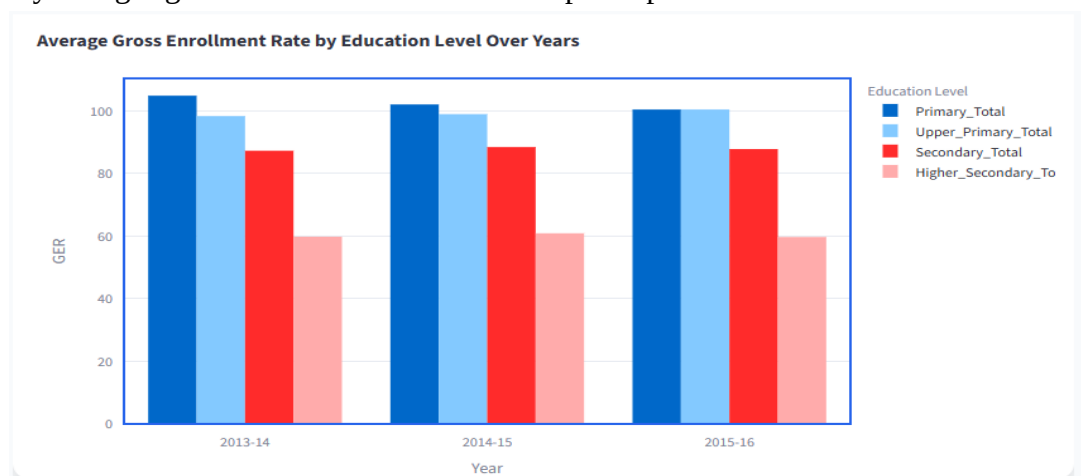
5.4 Results of Gross Enrollment Ratio (GER) Analysis

The Gross Enrollment Ratio (GER) analysis shows the participation of students in school education. The graphs compare enrollment levels across different educational stages and states.

The visualizations indicate that enrollment levels are not the same in every state. Some states show higher participation in education, while others have comparatively lower enrollment. These differences may be influenced by educational facilities, awareness, economic conditions, and government initiatives.

Observation:

- GER varies across different states.
- Some states have higher enrollment levels than others.
- The analysis highlights differences in educational participation.



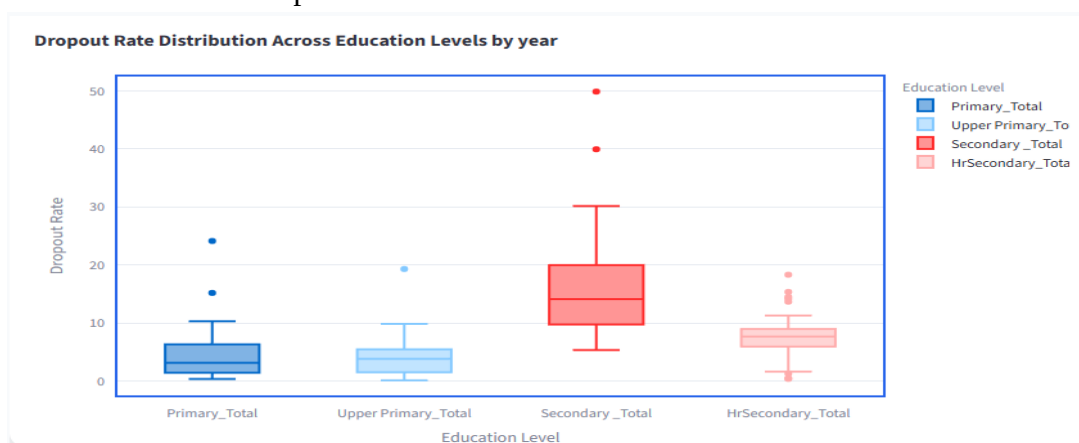
5.5 Results of Dropout Rate Analysis

The dropout rate analysis helps identify states where students leave school before completing their education. The graphs provide a comparison of dropout rates among different states and educational levels.

The analysis shows that dropout rates are not uniform across the country. While some states have relatively lower dropout rates, others require greater attention to improve student retention. Understanding these differences can help educational authorities identify areas that need improvement.

Observation:

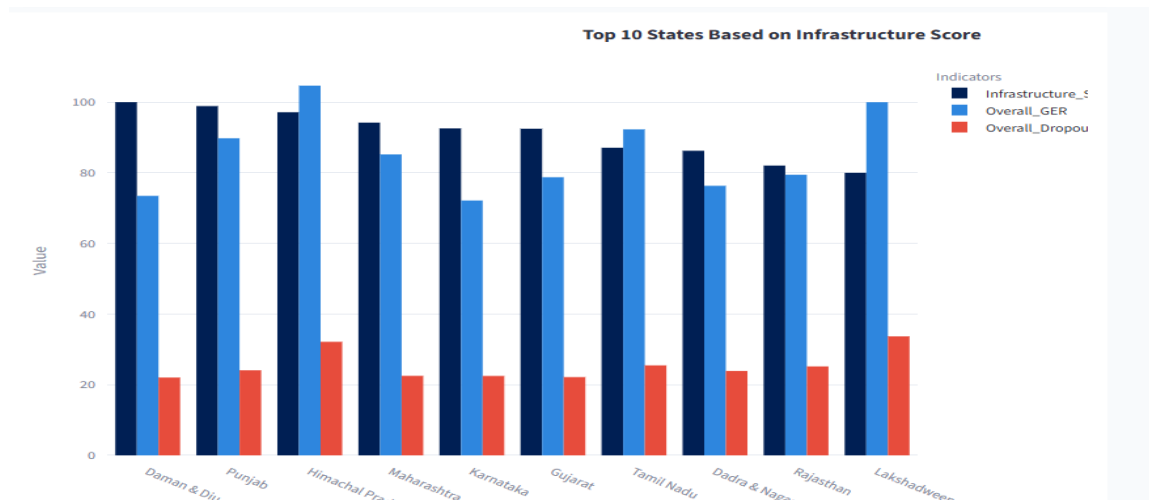
- Dropout rates vary among different states.
- Some regions require additional attention to reduce student dropouts.
- The visualization makes comparison easier.



5.6 Comparative Discussion

Comparing the different educational indicators provides a broader understanding of school performance. Infrastructure, enrollment, and dropout rates represent different aspects of the education system and together provide useful insights into educational development.

The dashboard allows users to compare these indicators through interactive visualizations. Such comparisons help identify educational strengths and areas where improvements may be required. Although each indicator provides different information, together they contribute to a more complete understanding of school education.



5.7 Key Findings

Based on the dashboard analysis, the following key findings were observed:

- Educational performance differs across Indian states.
- School infrastructure is not equally distributed among all states.
- Gross Enrollment Ratio varies from state to state.
- Dropout rates show noticeable differences across regions.
- Interactive visualizations make educational data easier to understand.
- Dashboard filters improve the exploration of educational information.
- State-wise comparisons help identify educational trends.
- Visual representation is more effective than numerical tables for understanding patterns.
- The dashboard provides a centralized platform for educational analysis.
- The project demonstrates the usefulness of data visualization in education.

5.8 Advantages of the Dashboard

The developed dashboard offers several advantages:

- Easy to use and navigate.
- Interactive visualizations improve understanding.
- State-wise comparisons are simple.
- Multiple educational indicators are available in one platform.
- Information is presented in a clear and organized manner.
- Users can quickly identify educational trends.
- Supports educational analysis and decision-making.

5.9 Limitations

Although the dashboard provides useful analysis, it has some limitations:

- The project is based on the available educational data.
- The dashboard focuses on descriptive analysis rather than prediction.
- The analysis is limited to the selected educational indicators.
- Future datasets may provide additional insights.

Chapter 6

Future Scope and Conclusion

6.1 Introduction

This chapter presents the overall conclusion of the **Indian School Performance Analysis** project. It summarizes the work carried out during the project and highlights the main outcomes achieved through data analysis and visualization. The chapter also discusses the limitations of the current project and suggests possible improvements that can be implemented in the future to make the system more effective and informative.

6.2 Conclusion

The **Indian School Performance Analysis** project was developed to analyze important educational indicators such as school infrastructure, Gross Enrollment Ratio (GER), and dropout rates across different states of India. The project successfully demonstrates how educational data can be organized and presented through an interactive dashboard using Python and Streamlit.

The dashboard provides a simple and user-friendly interface that allows users to explore educational information through different modules and visualizations. Instead of analyzing large tables of data, users can easily understand educational trends using graphs and summary statistics. This makes the analysis more meaningful and easier to interpret.

The project helps compare educational performance among different states and provides useful insights into school infrastructure, enrollment, and dropout patterns. It also shows how data visualization can simplify complex educational information and support better understanding of educational statistics.

Overall, this project has strengthened practical knowledge of Python programming, data preprocessing, data visualization, and dashboard development. It also demonstrates the importance of data analytics in solving real-world problems in the field of education. The developed dashboard serves as a useful tool for presenting educational data in a clear, interactive, and organized manner.

6.3 Achievements of the Project

The project achieved the following objectives:

- Developed an interactive dashboard using Streamlit.
- Analyzed educational data related to school infrastructure, Gross Enrollment Ratio (GER), and dropout rates.
- Presented educational information using interactive graphs and charts.
- Designed a user-friendly interface for easy navigation.
- Displayed educational statistics through summary cards and visualizations.
- Improved understanding of educational trends across different states.

- Applied data analysis and visualization techniques using Python.
- Enhanced practical knowledge of dashboard development and educational data analysis.

6.4 Limitations of the Project

Although the project provides useful educational analysis, it has some limitations.

- The project depends on the available educational dataset.
- The analysis is limited to the selected educational indicators.
- The dashboard provides descriptive analysis and does not predict future educational trends.
- State-level analysis is performed, while district-level analysis is not included.
- The dashboard focuses on visualization and comparison rather than advanced predictive modeling.

Despite these limitations, the project successfully achieves its objective of presenting educational data in an interactive and understandable manner.

6.5 Future Scope

The project can be further improved by adding more advanced features and using updated educational datasets. Future enhancements can increase the usefulness of the dashboard and provide deeper educational insights.

Possible future improvements include:

- Using the latest educational datasets for more up-to-date analysis.
- Extending the analysis to district or school level.
- Adding machine learning models to predict future dropout rates or enrollment trends.
- Including more educational indicators such as teacher availability, student performance, and attendance.
- Providing downloadable reports and data summaries.
- Developing a mobile-friendly version of the dashboard.
- Integrating real-time educational data when available.
- Adding more interactive filters and comparison options for users.
- Including AI-based recommendations to support educational planning and decision-making.

These improvements can make the dashboard more comprehensive and useful for students, researchers, educators, and policymakers.

References

1. <https://www.kaggle.com/datasets/vidyapb/indian-school-education-statistics>
2. <https://udiseplus.gov.in>
3. <https://docs.python.org/3/>
4. <https://pandas.pydata.org/docs/>
5. <https://numpy.org/doc/>
6. <https://plotly.com/python/>
7. <https://docs.streamlit.io/>
8. <https://code.visualstudio.com/docs>

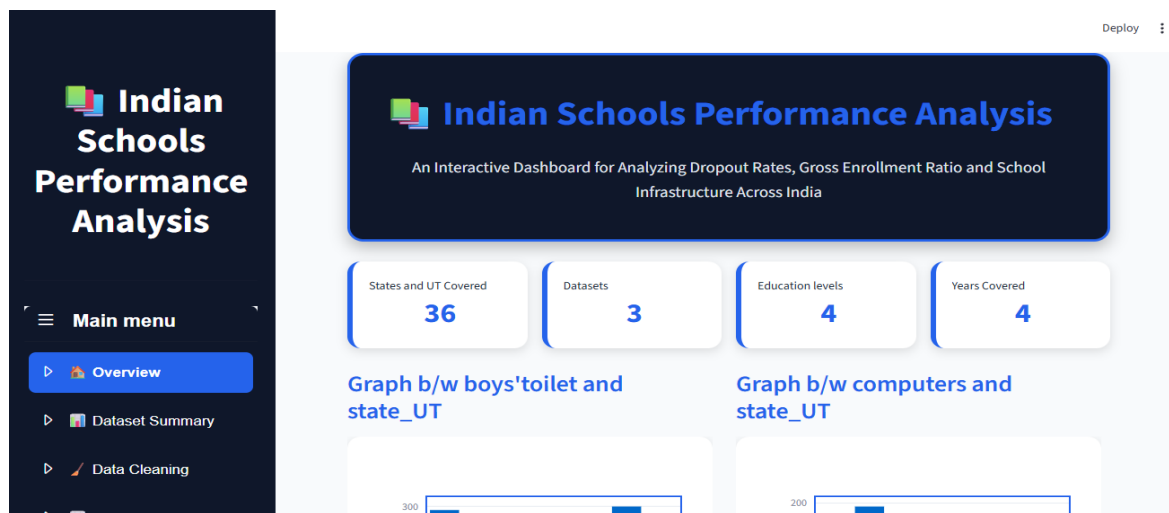
APPENDIX

Appendix A: Dashboard Screenshots

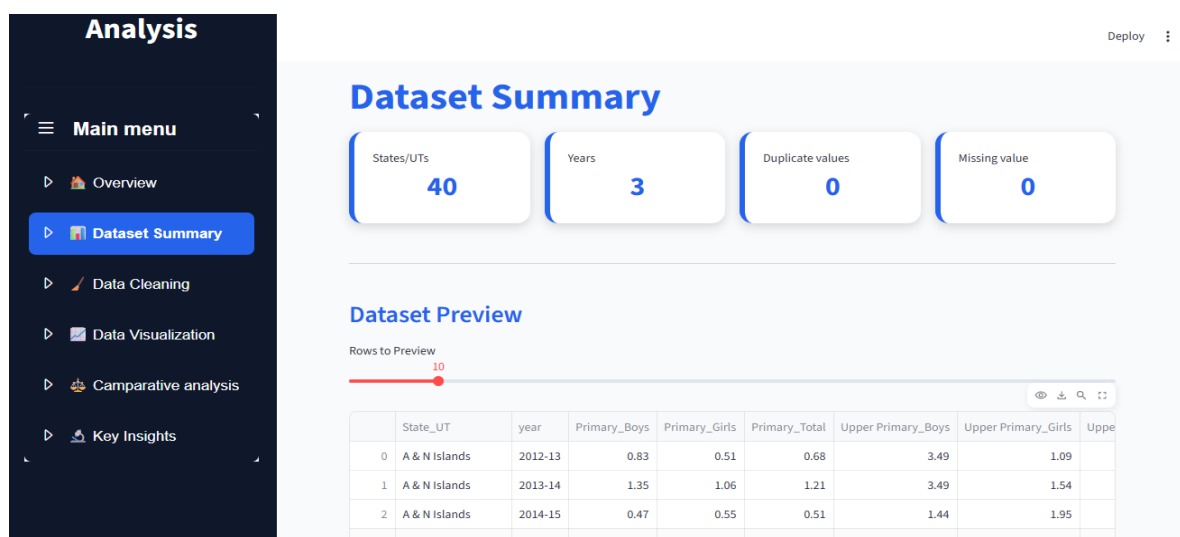
This appendix contains screenshots of the developed **Indian School Performance Analysis** dashboard. The screenshots provide an overview of the user interface and the different modules available in the dashboard.

Include the following screenshots:

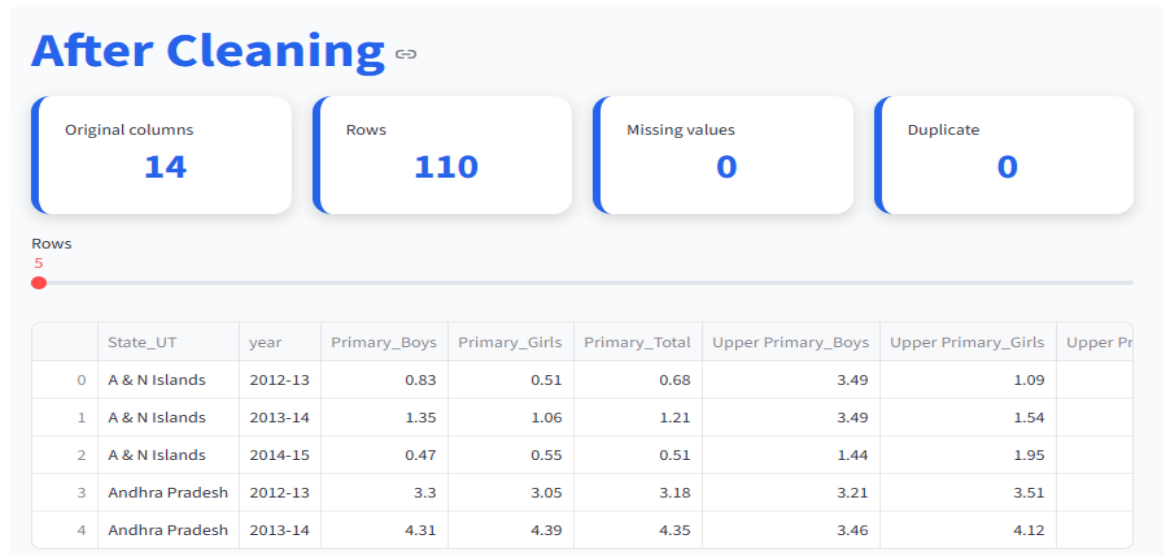
- Home Page



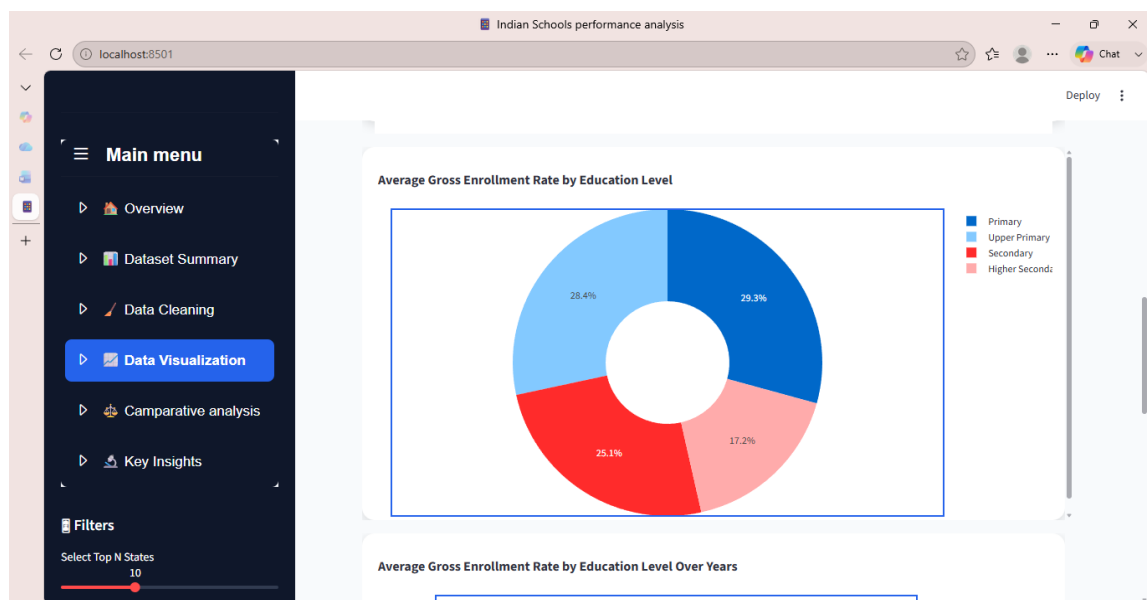
Data Summary



Data Processing



Data Visualization



Appendix B: Tools and Technologies Used

The project was developed using the following software and technologies:

Python – Programming language

Numpy - Numerical Operation

Pandas - Data Analysis

Plotly - Data Visualization

Streamlit - Dashboard Development

Visual Studio Code - Code editor

Appendix C: Project Workflow

The following workflow was followed during the development of the project:

